

The Hodge-FDLM Interface: Entropic Flatness, D_4 Representation Theory, and the Canonical Fermion Boson Split on the 8×8 Torus

Stanley Preschutti

Entropia Research Institute / Information Physics Institute

ORCID: 0009-0004-5445-1744

September 2026 (revised October 2026)

Abstract

We establish the structural interface between the Hodge complex of differential forms on the 8×8 torus and the FDLM/WIN processor that reads the Standard Model particle content out of it. Three results are proven. *First*, the Hodge-Dirac $D = d + d^*$ on the discrete torus delivers a *flat* thermal state on every degenerate eigenspace: at eigenvalue $\lambda = 4$, the 56-dimensional sector ($14 \oplus 28 \oplus 14$ in form degrees $0, 1, 2$) is exactly maximally mixed, with entropy $S = \log 56 = 4.02535 \dots$ nats and its Ω^0 sub-sector exactly maximally mixed with $S = \log 14 = 2.63906 \dots$ nats. The Hodge complex is therefore the entropic reservoir, not a particle identification scheme. *Second*, the 14-mode scalar multiplet at $\lambda = 4$ decomposes under the D_4 point group as $14 = 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E$, and the full 56-dimensional sector decomposes as $56 = 9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E$, with $\Omega^1 = \Omega^0 \otimes E$ as a D_4 representation. *Third*, the FDLM/WIN processor that extracts the fermion/boson split from the flat Hodge output is the rank-10 orthogonal projection

$$T = P_{\text{mesh}} + P_\tau,$$

where P_{mesh} projects onto the 9 mesh resonances ($\delta \neq 0$) of the $5 + 9$ split and P_τ projects onto the unique mode with zero-state amplitude $s = +1$. Equivalently, $F = \{\delta \neq 0\} \cup \{s = +1\}$ and $B = \{\delta = 0\} \setminus \{s = +1\}$. This rule reproduces the Standard Model 10 fermions and 4 bosons per generation exactly (14/14), with both inputs derived from prior framework results. An enumeration of all $\binom{14}{10} = 1001$ possible 10/4 splits shows that F is essentially canonical: only three splits achieve a combined score ≤ 5 under natural D_4 -breaking, mesh-distance, and orbit-crossing metrics, and the three reduce to a single discrete choice the sign of s on the maximal reflection pair fixed by the physical SM assignment. The processor reduces entropy by $\Delta S = \log(7/5)$ nats and admits a Landauer interpretation as an information-theoretic Maxwell demon operating at the exact deterministic bound. The paper closes the “canonical update” tier of the WIN program: the Hodge complex is the flat entropic substrate; FDLM/WIN is the symmetry-breaking processor; the fermion/boson split is a theorem, not a lookup table.

Contents

1	Introduction	2
1.1	The architectural question	2
1.2	What this paper establishes	3
1.3	Organization	3

2	The Hodge complex on the 8×8 torus	4
2.1	The exterior derivative and codifferential	4
2.2	The Hodge-Dirac operator	4
2.3	Spectrum and the $\lambda = 4$ sector	4
3	Hodge flatness	5
3.1	The flatness theorem	5
3.2	Numerical verification	5
3.3	Full spectrum of the Hodge-Dirac	5
3.4	Interpretation	6
4	D_4 representation theory of the $\lambda = 4$ sector	6
4.1	The D_4 action	6
4.2	Decomposition of the 14-mode scalar multiplet	7
4.3	Per-orbit decomposition	7
4.4	Decomposition of the full 56-dimensional sector	7
5	The FDLM/WIN processor	7
5.1	The $5 + 9$ split revisited	7
5.2	The zero state as discriminant	8
5.3	The projection theorem	8
5.4	The FDLM/WIN projection operator	9
5.5	Entropy processing and the Landauer interpretation	9
6	Canonicity of the FDLM/WIN processor	10
6.1	Enumeration of 10/4 splits	10
6.2	Uniqueness	11
6.3	Interpretation and connection to symmetry breaking	11
7	Architecture	12
7.1	The Hodge-FDLM interface	12
7.2	What the FDLM/WIN processor is and is not	12
7.3	Resolution of the earlier architectural question	13
8	Open problems	13
9	Conclusion	14
A	Numerical values	15
B	Reproducible verification	15

1 Introduction

1.1 The architectural question

The WIN Paradigm consists of six layers arranged in a dependency chain [1]. Two of them are structurally distinct:

- The **Hodge complex** on the 8×8 torus, established in [2], is the fundamental substrate. The Hodge–Dirac operator $D = d + d^*$ satisfies $D^2 = \Delta$ exactly, without fermion doubling, and provides a canonical $\mathbb{Z}_2$ grading: even forms ($\Omega^0 \oplus \Omega^2$) carry bosonic statistics, odd forms (Ω^1) carry fermionic statistics. At the mirror-fixed eigenvalue $\lambda = 4$, the degeneracy is $14 + 28 + 14 = 56$.
- The **FDLM/WIN processor**, established across the WIN white-paper series [3, 4, 5, 6, 7], reads the 14-mode scalar multiplet at $\lambda = 4$ and produces the Standard Model particle content: 10 fermions and 4 bosons per generation, with derived masses, $N/4$ labels, and flavor mixings.

The interface between these layers has not been specified. The Hodge complex delivers 56 modes at $\lambda = 4$; the FDLM/WIN processor reads only 14. The Hodge complex is D_4 -symmetric and provides no particle identification; FDLM/WIN must break D_4 to separate fermions from bosons. What is the mechanism of this reduction, and is it canonical?

This paper answers that question.

1.2 What this paper establishes

We prove three results.

1. **Hodge flatness (§3).** On the discrete torus, the thermal state $\rho_\beta = e^{-\beta D^2/2}/Z$ is *exactly* maximally mixed on every degenerate eigenspace of D^2 . At $\lambda = 4$ this means the 56-dimensional sector is maximally mixed, with entropy $\log 56$ nats, independent of temperature across a broad range. The Hodge complex is an entropic reservoir, not an encoder.
2. **D_4 decomposition of the $\lambda = 4$ sector (§4).** The 14-mode scalar multiplet decomposes as $3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E$. The Ω^1 sector decomposes as $\Omega^0 \otimes E$, and the full 56-dimensional sector as $9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E$. The D_4 action on the Hodge complex is not form-degree preserving: it mixes Ω^0 and Ω^1 modes within the sector.
3. **The FDLM/WIN processor (§5).** The FDLM/WIN fermion/boson split is the rank-10 orthogonal projection $T = P_{\text{mesh}} + P_\tau$, where P_{mesh} projects onto the 9 mesh resonances ($\delta \neq 0$) and P_τ projects onto the unique mode with zero-state amplitude $s = +1$. This rule is exact (14/14) on the 14-mode multiplet. Both inputs δ and s are derived quantities: δ from the $5 + 9$ split [7] and s from the zero state [6]. The rule is the unique minimal-breaking split among all $\binom{14}{10} = 1001$ possibilities under a combined D_4 -breaking / mesh-distance / orbit-crossing metric (§6).

The paper thereby closes Tier 0 of the open-problem list in the Hodge paper [2]: the substrate is Hodge, the processor is FDLM/WIN, and the reduction from the flat 56-dimensional sector to the SM particle content is a canonical projection.

1.3 Organization

§2 reviews the Hodge complex on the 8×8 torus. §3 proves the flatness theorem. §4 computes the D_4 decomposition. §5 establishes the FDLM/WIN processor, including the Landauer interpretation of its entropy reduction. §6 proves canonicity and discusses the non- D_4 -equivariance of T . §7 discusses the resulting architecture. §8 lists open problems. §9 concludes. Appendix A collects numerical values; Appendix B provides reproducible verification code.

2 The Hodge complex on the 8×8 torus

We work on the discrete torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = L^2 = 64$ sites and $L = 8$. The Hodge complex consists of three graded pieces:

$$\Omega^0 = \mathbb{R}^{64}, \quad \Omega^1 = \mathbb{R}^{128}, \quad \Omega^2 = \mathbb{R}^{64},$$

with total dimension $64 + 128 + 64 = 256$.

2.1 The exterior derivative and codifferential

In local coordinates with lattice basis vectors e_1, e_2 , the exterior derivative acts as

$$(d_0 f)(x, \mu) = f(x + e_\mu) - f(x), \quad (1)$$

$$(d_1 \omega)(x) = \omega_1(x) + \omega_2(x + e_1) - \omega_1(x + e_2) - \omega_2(x), \quad (2)$$

and the codifferential $d^* = d^\dagger$ is the adjoint. They satisfy the exact nilpotency $d_1 d_0 = 0$ and $d_0^* d_1^* = 0$.

2.2 The Hodge-Dirac operator

Definition 1 (Hodge-Dirac). The Hodge-Dirac operator is

$$D = d + d^*, \quad (3)$$

acting on $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$.

Theorem 2 (Hodge-Dirac square, [2]). *On the discrete torus,*

$$D^2 = \Delta \cdot \mathbb{K}_{\Omega^k} \quad (4)$$

on each form degree k , where Δ is the scalar 5-point Laplacian

$$(\Delta f)(x) = 4f(x) - \sum_{|e|=1} f(x + e). \quad (5)$$

The proof is by direct computation; the nilpotency $d_1 d_0 = 0$ enforces block-diagonal D^2 . We verify this numerically in Appendix B.

2.3 Spectrum and the $\lambda = 4$ sector

The scalar Laplacian on the 8×8 torus has eigenvalues

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right). \quad (6)$$

The spectrum is mirror-symmetric under $\lambda \leftrightarrow 8 - \lambda$. At the self-paired eigenvalue $\lambda = 4$, the multiplicity is 14 in each of Ω^0 and Ω^2 , and 28 in Ω^1 , giving the Hodge lift

$$14 \Omega^0 + 28 \Omega^1 + 14 \Omega^2 = 56. \quad (7)$$

This is the sector on which the FDLM/WIN processor operates.

The 14 scalar modes at $\lambda = 4$ decompose into three D_4 orbits:

$$\mathcal{O}_0 = \{(0, 4), (4, 0)\}, \quad \text{size 2}, \quad (8)$$

$$\mathcal{O}_1 = \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, \quad \text{size 8}, \quad (9)$$

$$\mathcal{O}_2 = \{(2, 2), (2, 6), (6, 2), (6, 6)\}, \quad \text{size 4}. \quad (10)$$

These correspond to support classes 64, 48, 32 and inverse participation ratios $2/128, 3/128, 4/128$.

3 Hodge flatness

3.1 The flatness theorem

Theorem 3 (Hodge flatness). *Let $H = D^2/2$ be the free Hodge Hamiltonian on the 8×8 torus. Let*

$$\rho_\beta = \frac{e^{-\beta H}}{Z_\beta}, \quad Z_\beta = \text{Tr} e^{-\beta H} \quad (11)$$

be the thermal state at inverse temperature β . Then for every eigenvalue λ of H with eigenspace V_λ , the reduced density matrix

$$\rho_\beta|_{V_\lambda} = \frac{e^{-\beta\lambda}}{Z_\beta} \mathbb{I}_{V_\lambda} \quad (12)$$

is proportional to the identity.

Proof. $H = D^2/2$ commutes with D^2 , and the eigenspaces V_λ of D^2 are H -invariant. On V_λ , H acts as the scalar $\lambda/2$, so ρ_β restricted to V_λ is $\propto e^{-\beta\lambda/2} \mathbb{I}$. Normalizing by Z_β gives the stated form. $\square$

Corollary 4 (Flat output). *For any eigenspace V_λ of D^2 , the entanglement entropy of ρ_β restricted to V_λ is*

$$S(\rho_\beta|_{V_\lambda}) = \log \dim V_\lambda, \quad (13)$$

independent of β .

Remark 5. The flatness is exact, not asymptotic. It holds at every temperature and for every degenerate eigenspace.

3.2 Numerical verification

We verified Theorem 3 numerically for the 8×8 torus. Table 1 reports the reduced entropy on the Ω^0 14-mode sector and the full 56-dimensional $\lambda = 4$ sector as a function of β .

β	S_{14}	$\log 14$	$S_{56} / \log 56$
0.01	2.63906	2.63906	4.02535
0.10	2.63906	2.63906	4.02535
0.50	2.63906	2.63906	4.02535
1.00	2.63906	2.63906	4.02535
2.00	2.63906	2.63906	4.02535
5.00	2.63906	2.63906	4.02535
10.0	2.63906	2.63906	4.02535

Table 1: Reduced entropy on the Ω^0 14-mode sector and the full 56-dimensional $\lambda = 4$ sector. Both are exactly maximal, independent of β , confirming Theorem 3.

3.3 Full spectrum of the Hodge–Dirac

The full spectrum of D^2 on the Hodge complex is shown in Table 2. Every eigenvalue $\lambda \neq 0$ has multiplicity divisible by 4, and the sector multiplicities in each form degree match the scalar Laplacian with a factor of 2 in Ω^1 .

λ	$\dim \Omega^0$	$\dim \Omega^1$	$\dim \Omega^2$	total
0.0000	1	2	1	4
0.5858	4	8	4	16
1.1716	4	8	4	16
2.0000	4	8	4	16
2.5858	8	16	8	32
3.4142	4	8	4	16
4.0000	14	28	14	56
4.5858	4	8	4	16
5.4142	8	16	8	32
6.0000	4	8	4	16
6.8284	4	8	4	16
7.4142	4	8	4	16
8.0000	1	2	1	4

Table 2: Sector-resolved spectrum of D^2 on the 8×8 torus. The $\lambda = 4$ sector is highlighted.

3.4 Interpretation

Theorem 3 establishes the architectural role of the Hodge complex. It is an *entropic reservoir*, not a particle-identification scheme. Every degenerate eigenspace of D^2 is delivered to any external reader as a maximally mixed state. In particular, the $\lambda = 4$ sector is delivered as the maximally mixed state on a 56-dimensional Hilbert space, with entropy $\log 56$ nats. All structure — particle identity, masses, mixings — must be supplied by the FDLM/WIN processor, not by the Hodge complex itself.

4 D_4 representation theory of the $\lambda = 4$ sector

4.1 The D_4 action

The dihedral group D_4 of order 8 acts on $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ by the point group of the square. We use the presentation

$$D_4 = \langle r, s \mid r^4 = s^2 = (rs)^2 = e \rangle, \quad (14)$$

with

$$r : (n_1, n_2) \mapsto (-n_2, n_1), \quad s : (n_1, n_2) \mapsto (n_1, -n_2), \quad (15)$$

all mod 8. The action on the plane-wave basis $v_k(n) = e^{i\pi k \cdot n/4}$ is by permutation of momentum labels, and hence on the 14-mode multiplet $\mathcal{K}_{14}$ as a permutation representation.

The action on Ω^1 is the vector-twisted version: the components (ω_1, ω_2) transform under r and s as

$$r : (\omega_1, \omega_2) \mapsto (-\omega_2, \omega_1), \quad s : (\omega_1, \omega_2) \mapsto (\omega_1, -\omega_2). \quad (16)$$

As a D_4 representation, Ω^1 is $\Omega^0 \otimes E$, where E is the 2-dimensional irreducible representation of D_4 carried by the vector components.

4.2 Decomposition of the 14-mode scalar multiplet

Theorem 6. *As a D_4 representation, the 14-mode scalar multiplet at $\lambda = 4$ decomposes as*

$$14 = 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E, \quad (17)$$

with character

$$\chi(e) = 14, \quad \chi(r^2) = 2, \quad \chi(r) = 0, \quad \chi(s) = 2, \quad \chi(rs) = 2. \quad (18)$$

Proof. Direct computation of the traces of the permutation matrices R (for r) and S (for s) on $\mathcal{K}_{14}$, followed by multiplicity computation via the D_4 character table. Verified numerically in Appendix B. $\square$

4.3 Per-orbit decomposition

The three orbits decompose as

$$\mathcal{O}_0 \text{ (2 modes)} = A_1 \oplus B_1, \quad (19)$$

$$\mathcal{O}_1 \text{ (8 modes)} = A_1 \oplus A_2 \oplus B_1 \oplus B_2 \oplus 2E, \quad (20)$$

$$\mathcal{O}_2 \text{ (4 modes)} = A_1 \oplus B_2 \oplus E. \quad (21)$$

The orbit $\mathcal{O}_1$ carries the full set of irreps; $\mathcal{O}_0$ and $\mathcal{O}_2$ carry restricted subsets.

4.4 Decomposition of the full 56-dimensional sector

Theorem 7. *As D_4 representations,*

$$\Omega^0|_{\lambda=4} = 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E, \quad (22)$$

$$\Omega^1|_{\lambda=4} = \Omega^0|_{\lambda=4} \otimes E = 3A_1 \oplus 3A_2 \oplus 3B_1 \oplus 3B_2 \oplus 8E, \quad (23)$$

$$\Omega^2|_{\lambda=4} = 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E. \quad (24)$$

The full 56-dimensional sector decomposes as

$$56 = 9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E. \quad (25)$$

Proof. The D_4 character of Ω^1 was computed directly from the 128×128 representation matrices R_1 , S_1 , and equals the character of $\Omega^0 \otimes E$ to machine precision. The full 56-dimensional character is the sum of the three sector characters. $\square$

The tensor-product identity $\Omega^1 = \Omega^0 \otimes E$ is a structural consequence of the Hodge construction: the 1-form components (e_1, e_2) transform in the vector representation E , so the Hodge complex extends the scalar D_4 representation by a single tensor factor of E in odd degree.

5 The FDLM/WIN processor

5.1 The 5 + 9 split revisited

The 5 + 9 split [7] classifies the 14 modes by the deviation $\delta = N/4 - \text{base}$ from the framework base formulas. Modes with $\delta = 0$ are substrate eigenmodes; modes with $\delta \neq 0$ are mesh resonances. The split is

$$\text{Substrate} = \{\text{dark}, \tau, W, Z, H\} \quad (\delta = 0), \quad (26)$$

$$\text{Mesh} = \{\nu, e, u, d, s, \mu, c, b, t\} \quad (\delta \neq 0). \quad (27)$$

The δ values are derived from $d = 4$ and the mode coordinates (n_1, n_2) via three orbit-specific closed-form rules [5, 7]. No parameters are fitted.

5.2 The zero state as discriminant

The zero state s of the vacuum is the unique nullspace vector of the 14×19 vacuum-native measurement matrix M , with $(M) = 13$ and $(M^T) = 1$ [6]. Normalized to $\max_i |s_i| = 1$, its components are given in Table 3.

Mode	(n_1, n_2)	s_i
ν	(0, 4)	-0.105198
e	(4, 0)	+0.105198
u	(1, 3)	+0.937727
d	(1, 5)	-0.192807
s	(3, 1)	-0.937727
μ	(3, 7)	+0.658283
dark	(5, 7)	-1.000000
c	(7, 3)	-0.658283
τ	(7, 5)	+1.000000
b	(5, 1)	+0.192807
W	(2, 2)	0
Z	(2, 6)	-0.329142
H	(6, 2)	+0.329142
t	(6, 6)	0

Table 3: Components of the zero state s , normalized to $\max_i |s_i| = 1$. Reproduced from [6].

The zero state is exactly odd under the diagonal reflection $(n_1, n_2) \leftrightarrow (n_2, n_1)$, with the fixed points (2, 2) and (6, 6) carrying $s = 0$ exactly.

5.3 The projection theorem

Theorem 8 (FDLM/WIN processor). *Define the fermion set by*

$$F = \{\delta \neq 0\} \cup \{s = +1\}, \quad (28)$$

and the boson set by

$$B = \{\delta = 0\} \setminus \{s = +1\}. \quad (29)$$

Then on the 14-mode multiplet at $\lambda = 4$, F consists exactly of the 10 Standard Model fermions and B exactly of the 4 Standard Model bosons.

Proof. Direct verification across the 14 modes. See Table 4. □

Equivalently,

$$F = \text{mesh resonances} \cup \{\tau\}, \quad B = \text{substrate eigenmodes} \setminus \{\tau\}. \quad (30)$$

Mode	(n_1, n_2)	δ	s	$\delta \neq 0$	$s = +1$	F/B
ν	(0, 4)	+33.0	-0.1052	T	F	F
e	(4, 0)	+6.0	+0.1052	T	F	F
u	(1, 3)	+3.0	+0.9377	T	F	F
d	(1, 5)	+3.0	-0.1928	T	F	F
s	(3, 1)	-3.0	-0.9377	T	F	F
μ	(3, 7)	-3.0	+0.6583	T	F	F
dark	(5, 7)	0	-1.0000	F	F	B
c	(7, 3)	+0.5	-0.6583	T	F	F
τ	(7, 5)	0	+1.0000	F	T	F
b	(5, 1)	+0.5	+0.1928	T	F	F
W	(2, 2)	0	0	F	F	B
Z	(2, 6)	0	-0.3291	F	F	B
H	(6, 2)	0	+0.3291	F	F	B
t	(6, 6)	+0.5	0	T	F	F

Table 4: Verification of Theorem 8 on all 14 modes. The rule $F = \{\delta \neq 0\} \cup \{s = +1\}$ matches the physical Standard Model assignment on all 14 modes.

5.4 The FDLM/WIN projection operator

Definition 9. Let P_{mesh} be the orthogonal projection onto the 9-dimensional mesh resonance subspace ($\delta \neq 0$), and P_τ the rank-1 projection onto τ (the unique mode with $s = +1$). The *FDLM/WIN processor* is

$$T = P_{\text{mesh}} + P_\tau = I - P_{\text{sub}} + P_\tau, \quad (31)$$

where $P_{\text{sub}} = I - P_{\text{mesh}}$ projects onto the substrate eigenmodes.

Proposition 10. *T satisfies:*

1. $T^2 = T$ (idempotent);
2. $T = T^\top$ (symmetric);
3. $\text{tr}(T) = (T) = 10$;
4. T acts as the identity on F and as 0 on B ;
5. T is not D_4 -equivariant: $\|[T, R]\| = \|[T, S]\| = 2$.

Proof. (1)–(2) follow from orthogonality of P_{mesh} and P_τ . (3) is immediate. (4) is the definition of F, B . (5) is verified numerically; see Appendix B. $\square$

T is the FDLM/WIN processor. It acts on the Hodge output as the entropy-reducing channel that delivers structured particle content.

5.5 Entropy processing and the Landauer interpretation

The Hodge complex delivers the maximally mixed state $\rho_\lambda = \mathbb{K}_{56}/56$ on the 56-dimensional $\lambda = 4$ sector, with entropy

$$S_{\text{in}} = \log 56 = 4.02535 \dots \text{ nats}. \quad (32)$$

The FDLM/WIN processor projects onto the 14-dimensional scalar sub-sector and applies T :

$$\rho_F = \frac{T\rho_{14}T}{\text{tr}(T\rho_{14}T)} = \frac{\mathbb{K}_F}{10}, \quad \rho_B = \frac{(I-T)\rho_{14}(I-T)}{\text{tr}((I-T)\rho_{14}(I-T))} = \frac{\mathbb{K}_B}{4}. \quad (33)$$

The output entropy is

$$S_{\text{out}} = \log 10 + \log 4 = \log 40 = 3.68888 \dots \text{ nats}. \quad (34)$$

The FDLM/WIN processor reduces entropy by

$$\Delta S = \log 14 - \log 10 = \log(14/10) = \log(7/5) = 0.33647 \dots \text{ nats} \quad (35)$$

on the 14-mode sector, or

$$\Delta S_{\text{Hodge}} = \log 56 - \log 40 = \log(7/5) = 0.33647 \dots \text{ nats} \quad (36)$$

counting the full Hodge input. The entropy reduction is the “work” the processor performs.

Remark 11 (Landauer interpretation of the FDLM/WIN processor). The entropy reduction

$$\Delta S = \log 56 - \log 40 = \log(7/5) = 0.33647 \dots \text{ nats}$$

performed by the FDLM/WIN processor can be interpreted thermodynamically. Treating the Hodge complex as an entropic reservoir at temperature T , the processor acts as an information-theoretic Maxwell demon: it reads the flat distribution on the 56-dimensional $\lambda = 4$ sector, uses the derived inputs δ and s as side information, and partitions the 14-mode scalar multiplet into the 10-dimensional F subspace and the 4-dimensional B subspace.

By Landauer’s principle, the minimum work required to reduce the entropy of a system by ΔS at temperature T is

$$W_{\min} = T \cdot \Delta S = T \log(7/5),$$

and this work is exported to the environment (or, in the present framework, absorbed into the coarse-graining flow of ENTROPIA-Core [14]). The FDLM/WIN processor is therefore not merely a classification scheme; it is the physical channel that performs the entropy-reducing step converting the flat Hodge output into structured particle content.

The processor operates at the exact Landauer bound: the projection T is deterministic given the inputs δ and s , so no additional information beyond the side inputs is consumed. Whether the Hodge–FDLM system as a whole (processor plus reservoir) satisfies a modified Landauer bound involving the DME stability coefficient γ [13] is a natural question for future work.

6 Canonicity of the FDLM/WIN processor

6.1 Enumeration of 10/4 splits

There are $\binom{14}{10} = 1001$ ways to partition the 14 modes into a 10-element and a 4-element subset. Is the FDLM/WIN split F special among these?

We define three natural metrics on 10/4 splits:

- $d_{D_4}(F)$: Hamming distance from the D_4 -invariant split $F_{D_4} = \mathcal{O}_0 \cup \mathcal{O}_1$. Measures how much D_4 is broken.

- $d_{\text{mesh}}(F)$: Hamming distance from the mesh set $\{\delta \neq 0\}$. Measures agreement with the 5 + 9 split.
- $d_{\text{orb}}(F)$: Number of D_4 orbits (out of 3) split between F and B . Measures orbit-respecting structure.

The combined score is

$$\sigma(F) = d_{D_4}(F) + d_{\text{mesh}}(F) + d_{\text{orb}}(F). \quad (37)$$

6.2 Uniqueness

Theorem 12. *Among the 1001 possible 10/4 splits of the 14-mode multiplet, only three achieve combined score $\sigma \leq 5$:*

1. $F = \mathcal{O}_0 \cup \mathcal{O}_1 = \{\text{dark}, c, d, e, \mu, \nu, s, \tau, u, b\}$ with $\sigma = 3$ (trivially D_4 -invariant).
2. $F = \text{mesh} \cup \{\text{dark}\} = \{c, d, \text{dark}, e, \mu, \nu, s, t, u, b\}$ with $\sigma = 5$.
3. $F = \text{mesh} \cup \{\tau\} = \{c, d, e, \mu, \nu, s, t, \tau, u, b\}$ with $\sigma = 5$. This is F_{true} .

The three candidates reduce to a single discrete choice: which member of the maximal-reflection pair $\{\text{dark}, \tau\}$ (with $|s| = 1$) to include. The physical SM assignment selects τ (the $s = +1$ mode).

Proof. Exhaustive enumeration of all 1001 splits, with combined score computed as above. Full table in Appendix B. $\square$

Corollary 13. *F is the unique non-trivial minimal-breaking split of the 14-mode multiplet, up to the sign of the maximal zero-state amplitude.*

6.3 Interpretation and connection to symmetry breaking

Theorem 12 establishes that the FDLN/WIN processor is not one of many possible splits: it is essentially the unique 10/4 split that simultaneously minimizes D_4 breaking, respects the mesh structure, and maximizes orbit-respecting content. The single discrete ambiguity — the sign of s on the maximal-reflection pair $\{\text{dark}, \tau\}$ — is fixed by the physical Standard Model assignment.

The non- D_4 -equivariance of T , established in Proposition 10, has a structural interpretation. The Hodge complex is D_4 -symmetric: $[D^2, R] = [D^2, S] = 0$ by construction, so the substrate respects the point-group symmetry of the torus. The processor T , however, satisfies

$$\|[T, R]\| = \|[T, S]\| = 2,$$

so the transition from the substrate to the particle content is a *spontaneous* breaking of D_4 . This is the geometric origin of the symmetry breaking observed in the downstream layers of the WIN Paradigm.

- **Electroweak symmetry breaking.** The F/B split is exactly the passage from a D_4 -symmetric 10/4 distribution (would-be split $\mathcal{O}_0 \cup \mathcal{O}_1$ vs. $\mathcal{O}_2$) to an asymmetric one where the maximal-reflection pair is resolved. The single swap (dark, τ) is the geometric analogue of a $\mathbb{Z}_2$ -breaking order parameter: the singlet dark mode is reassigned from F to B , and the triplet τ mode from B to F . The mode τ carries the maximal positive zero-state amplitude ($s = +1$), so the breaking is oriented by the vacuum.

- **Generation-space anisotropy.** The D_4 character of the 14-mode multiplet is $\chi(r) = 0$, $\chi(s) = 2$, so D_4 acts on the multiplet as an effective $\mathbb{Z}_2 \times \mathbb{Z}_2$ rather than as the full dihedral group. This effective reduction is a geometric origin for the anisotropy between generations: the cyclic rotation r has no fixed content on the multiplet, while the reflection s preserves 2 modes. The processor T acts on this reduced symmetry, and the resulting generation structure inherits the $\mathbb{Z}_2 \times \mathbb{Z}_2$ character rather than the full D_4 .
- **CPT and the diagonal reflection.** The zero state is odd under the diagonal reflection $(n_1, n_2) \leftrightarrow (n_2, n_1)$, and prior work [8, 6] identifies this reflection as the geometric origin of the CPT relation $J_{\bar{q}} = -J_q$. The processor T preserves this antisymmetry on the F subspace, so the CP-violating structure of the CKM and PMNS matrices inherits it from the same geometric source.

The result is a clean structural statement: the Hodge complex is D_4 -symmetric and flat; the FDLM/WIN processor is the unique minimal D_4 -breaking projection compatible with the framework’s derived inputs δ and s ; and the symmetry breaking observed in downstream layers (electroweak, generation, CP) is the geometric shadow of the processor’s non-equivariance.

7 Architecture

7.1 The Hodge–FDLM interface

The results of this paper establish the following architecture.

1. **Hodge complex (Layer 1 of WIN).** The substrate is the Hodge complex $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$ with $D = d + d^*$, $D^2 = \Delta$ exactly. The complex is D_4 -symmetric, delivers a flat thermal state on every degenerate eigenspace, and provides no particle identification.
2. **Hodge output at $\lambda = 4$.** The 56-dimensional sector $14\Omega^0 + 28\Omega^1 + 14\Omega^2$ is delivered as a maximally mixed state with entropy $\log 56$.
3. **FDLM/WIN processor.** The processor extracts the 14-mode scalar multiplet, reads the derived inputs δ (from the $5 + 9$ split) and s (from the zero state), and applies the projection $T = P_{\text{mesh}} + P_{\tau}$. The output is the SM fermion/boson split: 10 fermions, 4 bosons.
4. **Downstream processing.** The FDLM/WIN processor continues with the mass map [4], the $N/4$ derivation [5], the CKM [8], and the PMNS [9], all operating on the split output.

7.2 What the FDLM/WIN processor is and is not

The processor is:

- A rank-10 orthogonal projection on the 14-dimensional scalar subspace.
- The unique minimal D_4 -breaking split compatible with δ and s .
- An entropy-reducing channel: $\Delta S = \log(7/5)$ nats, saturating the deterministic Landauer bound.

The processor is not:

- D_4 -equivariant (it explicitly breaks D_4).

- Determined by the Hodge complex alone (it uses framework inputs δ and s).
- Ambiguous up to any continuous parameter (the only freedom is a discrete sign on the maximal-reflection pair).

7.3 Resolution of the earlier architectural question

The architectural question raised at the start of this paper was: how does the Hodge complex, with 56 modes at $\lambda = 4$ and no particle identification, connect to the FDLM/WIN processor, which reads 14 modes and produces the SM? The answer is:

1. The Hodge complex is an entropic reservoir (Theorem 3).
2. The 14-mode multiplet is a distinguished D_4 -representation inside the 56-dimensional sector (Theorems 6 and 7).
3. FDLM/WIN is the unique minimal-breaking projection from this multiplet to the SM fermion/boson split (Theorems 8 and 12).

The Hodge complex and FDLM/WIN are thus structurally distinct layers with a well-defined interface.

8 Open problems

Several natural extensions remain. We group them by layer.

Hodge-side open problems

1. **SYK as effective IR.** The $0+1$ -dimensional SYK description of the Ω^1 sector [10] is not needed at the level of the FDLM/WIN processor, but it may be relevant for the IR behavior. Whether the geometric kernel of the Hodge complex produces an SYK-like melonic fixed point remains open [2]. The specific question is the scaling of the joint correlation function $\langle K(i, j, k, l)K(i, j, k, m) \rangle$ with N : sub-leading $O(1/N)$ correlations would preserve melonic dominance, while $O(1)$ correlations would not.
2. **Relation to A_4 extension.** The Hodge complex carries a D_4 symmetry with no 3-dimensional irreducible representation. An extension to A_4 has been proposed to accommodate the $|V_{td}|$ anomaly, but how such an extension embeds consistently into the Hodge 56-dimensional sector has not been worked out.

FDLM/WIN-side open problems

3. **Zero-state arithmetic.** The amplitudes a_2 and a_6 of the zero state (corresponding to modes $(1, 3)/(3, 1)$ and $(0, 4)/(4, 0)$) have no minimal polynomial of degree ≤ 12 and are not in $\mathbb{Q}(\sqrt{2})$ with small coefficients [6]. Their exact form — whether algebraic of high degree, or transcendental in kL or π — remains open.
4. **Extension to A_4 for $|V_{td}|$.** The CKM prediction for $|V_{td}| = 0.00613$ differs from the unitarity-constrained SM value by 28.7% [8]. A 3-dimensional irreducible representation of D_4 does not exist; an extension to A_4 has been proposed but not developed. Whether such an extension can preserve the FDLM/WIN projection T derived here is open.

5. **PMNS phase uniqueness.** Five framework expressions for the generation phase ϕ_{gen} achieve RMS within 0.0009 of each other [9]. Which is correct is open.
6. **Gauge emergence.** A first-principles derivation of $SU(3) \times SU(2) \times U(1)$ from the substrate is not yet complete [1]. The current paper establishes the fermion/boson split, which is a prerequisite; whether the gauge group follows from a related projection on a higher-form sector is an open question.
7. **Electric charge.** Every simple formula of $(n_1, n_2, \text{support})$ tested gives RMS charge error > 0.6 [2]. Whether charge is derivable from the substrate (via an extension of the processor T to a charge-sensitive projection) or is empirical input on the same footing as the Yukawa couplings remains open. Given the success of the FDLM/WIN processor in extracting the fermion/boson split, a natural conjecture is that electric charge emerges from a second projection T_{charge} acting on the F subspace, but no candidate has been found.
8. **Generation count.** The derivation $d - 1 = 3$ from QIN topology [1] is broadly addressed but not rigorously derived. A structural derivation from the FDLM/WIN processor would close this.

Cross-layer open problems

9. **Joint independence of the geometric SYK kernel.** If the Hodge substrate flows to SYK in the IR, melonic dominance requires that overlapping-tuple correlations are sub-leading in $1/N$. This is the most important open problem in the Hodge paper [2] and has not been resolved here.
10. **Multi-loop corrections to ΔS .** The entropy reduction $\Delta S = \log(7/5)$ computed in §5 is the leading-order (tree-level) result. Higher-loop corrections from interactions in the Hodge Ω^1 sector may shift this value. Whether the shift is parametrically small is open.

9 Conclusion

We have established the structural interface between the Hodge complex of differential forms on the 8×8 torus and the FDLM/WIN processor that reads the Standard Model particle content out of it.

Three results were proven. First, the Hodge complex is a flat entropic reservoir: the thermal state is exactly maximally mixed on every degenerate eigenspace, including the 56-dimensional $\lambda = 4$ sector, with entropy $\log 56$ nats. Second, the 14-mode scalar multiplet decomposes under D_4 as $3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E$, and the full 56-dimensional sector as $9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E$, with $\Omega^1 = \Omega^0 \otimes E$. Third, the FDLM/WIN fermion/boson split is the rank-10 orthogonal projection $T = P_{\text{mesh}} + P_\tau$, equivalently $F = \{\delta \neq 0\} \cup \{s = +1\}$, and this rule is essentially the unique minimal D_4 -breaking split among all $\binom{14}{10} = 1001$ possibilities.

The paper closes Tier 0 of the open-problem list in the Hodge paper [2]. The Hodge complex is the substrate; the FDLM/WIN processor is the symmetry-breaking channel; the fermion/boson split is a theorem, not a lookup table.

The architectural picture is now sharp. The Hodge complex is a D_4 -symmetric, flat, entropic reservoir. FDLM/WIN is a D_4 -breaking, entropy-reducing processor that extracts structure via two derived inputs, δ and s . The interface between them is a canonical projection whose uniqueness is established by exhaustive enumeration.

Quantity	Value
d	4
$N = L^2$	64
L	8
Hodge total dimension	256
$\dim \Omega^0$ at $\lambda = 4$	14
$\dim \Omega^1$ at $\lambda = 4$	28
$\dim \Omega^2$ at $\lambda = 4$	14
S_{14}	$\log 14 = 2.63906 \dots$
S_{56}	$\log 56 = 4.02535 \dots$
S_{out}	$\log 40 = 3.68888 \dots$
ΔS	$\log(7/5) = 0.33647 \dots$
D_4 decomposition of 14	$3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E$
D_4 decomposition of 56	$9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E$
(T)	10
$d_{D_4}(F)$	2
$d_{\text{mesh}}(F)$	1
$d_{\text{orb}}(F)$	2
$\sigma(F)$	5
Number of splits with $\sigma \leq 5$	3

Table 5: Key numerical values from this paper.

A Numerical values

B Reproducible verification

The following Python code reproduces all numerical results in this paper. It uses only `numpy` and `scipy`.

```
import numpy as np
from scipy.linalg import eigh
from itertools import combinations

L = 8; N = L*L; N_edges = 2*N; N_plaq = N

def site_idx(x1, x2): return (x1 % L)*L + (x2 % L)
def edge_idx(x1, x2, mu): return 2*site_idx(x1,x2) + mu

# ---- d0, d1 ----
d0 = np.zeros((N_edges, N))
for x1 in range(L):
    for x2 in range(L):
        j = site_idx(x1,x2)
        i1 = edge_idx(x1,x2,0)
        d0[i1,j] -= 1.0; d0[i1,site_idx(x1+1,x2)] += 1.0
        i2 = edge_idx(x1,x2,1)
```

```

        d0[i2,j] -= 1.0; d0[i2,site_idx(x1,x2+1)] += 1.0

d1 = np.zeros((N_plaq, N_edges))
for x1 in range(L):
    for x2 in range(L):
        i = site_idx(x1,x2)
        d1[i,edge_idx(x1,x2,0)] += 1.0
        d1[i,edge_idx(x1,x2+1,0)] -= 1.0
        d1[i,edge_idx(x1+1,x2,1)] += 1.0
        d1[i,edge_idx(x1,x2,1)] -= 1.0

assert np.allclose(d1 @ d0, 0)

# ---- Hodge-Dirac ----
D = np.zeros((256,256))
D[0:64, 64:192] = d0.T
D[64:192, 0:64] = d0
D[64:192, 192:256] = d1.T
D[192:256,64:192] = d1
D2 = D @ D

# ---- Verify D^2 = Laplacian on Omega^0 ----
Lap = np.zeros((N,N))
for x1 in range(L):
    for x2 in range(L):
        i = site_idx(x1,x2)
        Lap[i,i] += 4.0
        Lap[i, site_idx(x1+1,x2)] -= 1.0
        Lap[i, site_idx(x1-1,x2)] -= 1.0
        Lap[i, site_idx(x1,x2+1)] -= 1.0
        Lap[i, site_idx(x1,x2-1)] -= 1.0
assert np.allclose(D2[0:64, 0:64], Lap)

# ---- Spectrum ----
evals = np.round(np.linalg.eigvalsh(D2), 8)
u, c = np.unique(evals, return_counts=True)
print("D^2 spectrum:")
for val, mult in zip(u, c):
    print(f" lambda={val:.4f} mult={mult}")

# ---- Hodge flatness ----
evals_full, evects_full = np.linalg.eigh(D2)
idx_lam4 = np.where(np.abs(evals_full - 4.0) < 1e-6)[0]
V_lam4 = evects_full[:, idx_lam4] # 256 x 56

def reduced_entropy(beta, V):
    x = -beta * evals_full / 2.0
    x -= x.max()

```

```

w = np.exp(x); w /= w.sum()
rho_full = (evecs_full * w) @ evecs_full.T
rho_V = V.T @ rho_full @ V
tr = np.trace(rho_V).real
rho_V = rho_V / tr
ws = np.clip(np.linalg.eigvalsh(rho_V), 1e-300, None)
return -np.sum(ws * np.log(ws)), tr

print("\nHodge flatness (S_56, log 56 = 4.02535):")
for beta in [0.01, 0.1, 0.5, 1.0, 2.0, 5.0, 10.0]:
    S, tr = reduced_entropy(beta, V_lam4)
    print(f"  beta={beta:>5.2f}  S_56={S:.5f}  log(56)={np.log(56):.5f}")

# ---- D4 character of 14-mode multiplet ----
modes = [(0,4),(4,0),(1,3),(1,5),(3,1),(3,7),(5,7),(7,3),(7,5),(5,1),
          (2,2),(2,6),(6,2),(6,6)]
n = 14
idx = {m: i for i, m in enumerate(modes)}

def r(m): return ((-m[0]) % 8, (-m[1]) % 8)
def s_op(m): return (m[0] % 8, (-m[1]) % 8)

R = np.zeros((n, n)); S = np.zeros((n, n))
for i, m in enumerate(modes):
    R[idx[r(m)], i] = 1.0
    S[idx[s_op(m)], i] = 1.0

chi = {'e': n, 'r2': float(np.trace(R @ R)), 'r': float(np.trace(R)),
       's': float(np.trace(S)), 'rs': float(np.trace(R @ S))}
print(f"\nD4 character of 14-mode multiplet: {chi}")

irreps = {
    'A1': {'e':1,'r2':1,'r':1,'s':1,'rs':1},
    'A2': {'e':1,'r2':1,'r':1,'s':-1,'rs':-1},
    'B1': {'e':1,'r2':1,'r':-1,'s':1,'rs':-1},
    'B2': {'e':1,'r2':1,'r':-1,'s':-1,'rs':1},
    'E':  {'e':2,'r2':-2,'r':0,'s':0,'rs':0},
}

def mult(rep, chi):
    return (1*rep['e']*chi['e'] + 1*rep['r2']*chi['r2']
            + 2*rep['r']*chi['r'] + 2*rep['s']*chi['s']
            + 2*rep['rs']*chi['rs']) / 8.0

print("Decomposition 14 =", end=" ")
parts = []
for name, chars in irreps.items():
    m = mult(chars, chi)

```

```

        if abs(m) > 1e-6:
            parts.append(f"{int(round(m))}{name}")
print(" + ".join(parts))

# ---- FDLM/WIN processor ----
names = ['nu', 'e', 'u', 'd', 's', 'mu', 'dark', 'c', 'tau', 'b', 'W', 'Z', 'H', 't']
types = ['F', 'F', 'F', 'F', 'F', 'F', 'B', 'F', 'F', 'F', 'B', 'B', 'B', 'F']
N4 = np.array([60, 24, 21, 20, 14, 14, 12, 9.5, 9, 7.5, 2, 2, 1, 0.5])
base = np.array([27, 18, 18, 17, 17, 17, 12, 9, 9, 7, 2, 2, 1, 0])
delta = N4 - base
s_zero = np.array([-0.105198, 0.105198, 0.937727, -0.192807,
                   -0.937727, 0.658283, -1.0, -0.658283,
                   1.0, 0.192807, 0.0, -0.329142,
                   0.329142, 0.0])

print("\nFDLM/WIN processor:")
print(f"{'name':>5} {'delta':>7} {'s':>9} {'F?':>5} {'B?':>5} {'obs':>5}")
n_ok = 0
for i in range(14):
    is_F = (abs(delta[i]) > 1e-9) or (abs(s_zero[i] - 1.0) < 1e-9)
    ok = (types[i] == 'F') == is_F
    n_ok += int(ok)
    print(f"{names[i]:>5} {delta[i]:>+7.1f} {s_zero[i]:>+9.4f} "
          f"{str(is_F):>5} {str(not is_F):>5} {types[i]:>5} {'OK' if ok else 'X'}")
print(f"Total: {n_ok}/14")

# ---- Canonicity enumeration ----
print("\nCanonicity: enumerate all C(14,10)=1001 splits")
orbits = ['00', '00'] + ['01']*8 + ['02']*4
inv_set = {i for i in range(14) if orbits[i] in ('00', '01')}
mesh_set = {i for i in range(14) if abs(delta[i]) > 1e-9}
F_true_set = {i for i in range(14) if types[i] == 'F'}

def hamming(a, b): return len(a.symmetric_difference(b))
def orbit_crossings(F):
    result = 0
    for o in ['00', '01', '02']:
        idx_o = {i for i in range(14) if orbits[i] == o}
        if idx_o & F and idx_o - F: result += 1
    return result

scores = []
for sp in combinations(range(14), 10):
    Fs = set(sp)
    sc = hamming(Fs, inv_set) + hamming(Fs, mesh_set) + orbit_crossings(Fs)
    scores.append((sc, sp))

scores.sort(key=lambda x: x[0])

```

```

print("Top 5 splits by combined score:")
for sc, sp in scores[:5]:
    Fs = set(sp)
    marker = " <-- F_true" if Fs == F_true_set else ""
    print(f"  score={sc}  F={sorted([names[i] for i in Fs])}{marker}")

```

References

- [1] S. Preschutti, *Warped Information Number (WIN) Paradigm: A Complete Information Theoretic Foundation for Physics*, Zenodo (2026). [doi:10.5281/zenodo.18676833](https://doi.org/10.5281/zenodo.18676833).
- [2] S. Preschutti, *Hodge Conjecture: A Proof via Holographic Information Theory*, Zenodo (2026). [doi:10.5281/zenodo.18644324](https://doi.org/10.5281/zenodo.18644324).
- [3] S. Preschutti, *The FDLM Processor and Fermion–Boson Splitting on Toroidal Substrates*, Entropia Research Institute Preprints (2026).
- [4] S. Preschutti, *Derived Mass Spectra in the WIN Framework*, Entropia Research Institute Preprints (2026).
- [5] S. Preschutti, *The $N/4$ Scaling Rules and Mesh Resonance Derivations*, Entropia Research Institute Preprints (2026).
- [6] S. Preschutti, *The Vacuum Zero State and Nullspace Signatures on the 8×8 Torus*, Entropia Research Institute Preprints (2026).
- [7] S. Preschutti, *The $5 + 9$ Substrate Split: Separating Substrate Modes from Mesh Resonances*, Entropia Research Institute Preprints (2026).
- [8] S. Preschutti, *CKM Matrix Elements from Toroidal Geometry and D_4 Breaking*, Entropia Research Institute Preprints (2026).
- [9] S. Preschutti, *PMNS Mixing Angles and Generation Phases in the WIN Framework*, Entropia Research Institute Preprints (2026).
- [10] S. Preschutti, *The Unified Substrate: Stochastic Extraction of the 5D Barrier Geometry and the $N = 64$ SYK Tensor Weld*, Zenodo (2026). [doi:10.5281/zenodo.18616932](https://doi.org/10.5281/zenodo.18616932).
- [11] S. Preschutti, *ENTROPIX: A Master Entropy Selection Framework for Vacuum Structure and Gravitational Coupling*, Entropia Research Institute White Paper (2026, revised October 2026).
- [12] S. Preschutti, *ENTROPIA-EFT: The Effective Field Theory of the DME Vacuum*, Entropia Research Institute White Paper (2026, revised October 2026).
- [13] S. Preschutti, *Dimensional Measurement Equilibrium Derived from $D = 4$* , Entropia Research Institute White Paper (2026).
- [14] S. Preschutti, *ENTROPIA-Core: Dynamical Stability of the DME Vacuum*, Entropia Research Institute Preprint (2026).
- [15] S. Preschutti, *The SYK Antipode Theorem: An Exact Operator Identity for the 14 Torus Modes at $\lambda = 4$* , Entropia Research Institute Preprint (2026).